



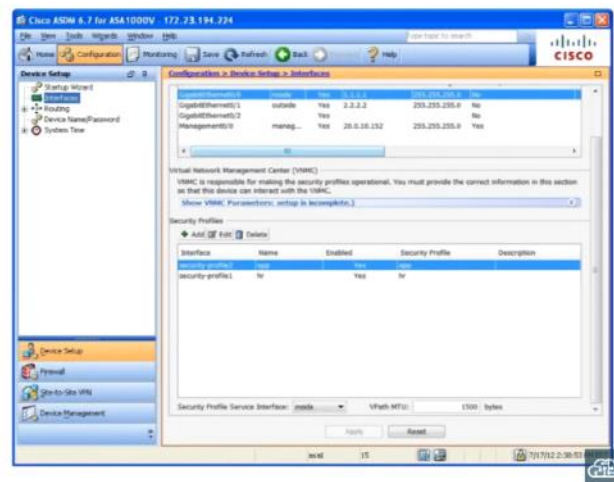
What is a CLI?

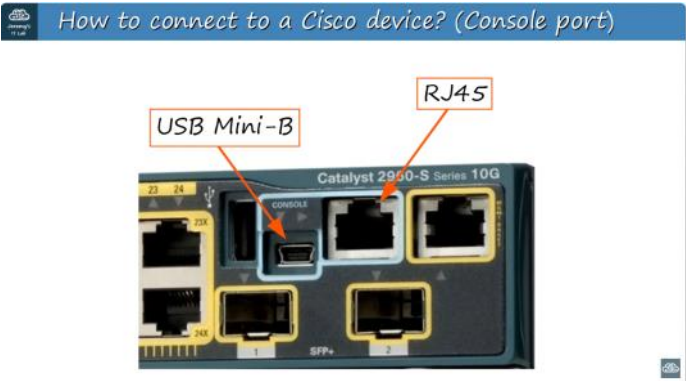
- Command-line interface
- The interface you use to configure Cisco devices

```
logging synchronous
stopbits 1
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
login
transport input all
!
end

R1(config)#int
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#no shutdown
R1(config-if)#
*Oct 27 00:35:00.987: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Oct 27 00:35:01.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config-if)#ip add
R1(config-if)#ip address 172.16.1.10 255.255.255.0
R1(config-if)#exit
R1(config)#
```

- GUI (Graphical User Interface)

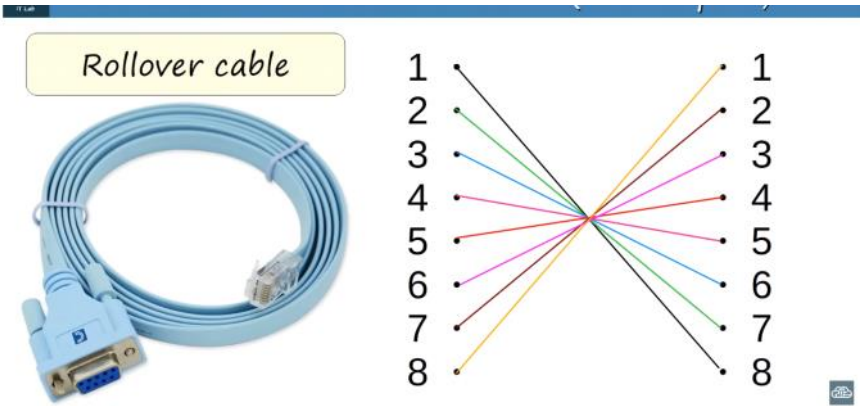


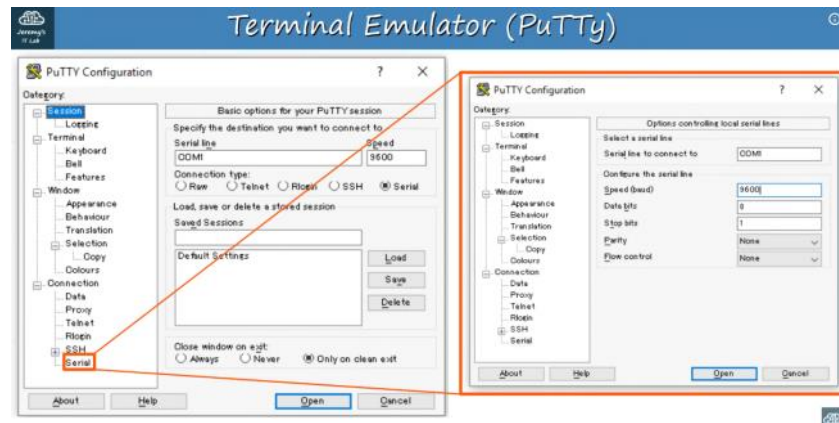
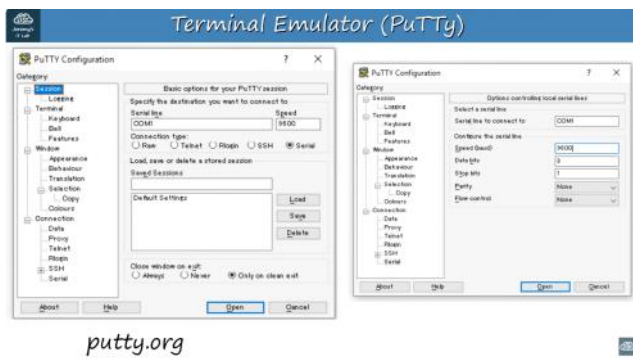


DB-9



- Rollover Cable (Console Cable)
- Also known as a rollover cable.
 - It is a console cable used to connect a computer terminal to a network device's console port.
 - This connection allows configuration and management of the network device.





Once you plug (Console port) into the device how do you access the device?
You use Putty

Serial Communication Settings

1) Speed (Baud Rate)

- The **Speed** box allows you to choose the speed (or **baud rate**) at which data is transmitted over the serial line.
- Common values:
 - 9600
 - 19200
 - 38400
 - 57600
- The correct speed depends on the device at the other end of the serial cable.

2) Data Bits

- The **Data Bits** option defines how many bits of actual data are transmitted in each byte.
- Typical values:
 - 7 bits
 - 8 bits (most common)

3) Stop Bits

- The **Stop Bits** setting determines how many stop bits are used to signal the end of each data packet.
- Typical values:
 - 1
 - 1.5
 - 2

4) Parity

- Parity** is a simple method used for **error checking**.

Types of parity:

- None:** No parity bit is used.
- Odd:** A parity bit is added so the total number of 1s is **odd**.
- Even:** A parity bit is added so the total number of 1s is **even**.
- Mark:** Parity bit is always set to 1.
- Space:** Parity bit is always set to 0.

5) Flow Control

- Flow control** manages how data is transmitted between devices.
- Default setting: **None**
 - Data is sent continuously.
 - No hardware or software control signals are used.
 - Data may be lost if one device sends faster than the other can handle.

Types of flow control:

- None:** No flow control.
- XON/XOFF (Software):**
 - Uses special characters to start or stop data transmission.
- RTS/CTS (Hardware):**
 - Uses dedicated wires for controlling data flow.
- DSR/DTR (Hardware):**
 - Uses DSR and DTR control lines for flow management.

Cisco CLI

15 February 2026 15:41



Cisco IOS CLI

```
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>
```

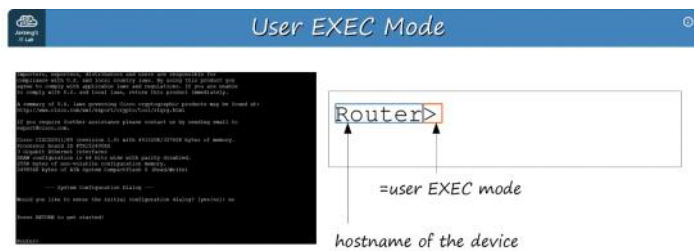
When you first boot up a Cisco device, it displays the following message:

Would you like to enter the initial configuration dialog? [yes/no]:

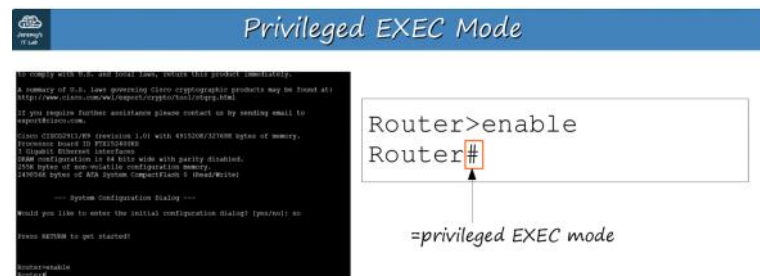
Press **no**.

If you type **no**, the device will take you directly to the **CLI** (Command Line Interface).

Then press **Enter** to continue.



- User EXEC mode is very limited.
- Users can look at some things, but can't make any changes to the configuration.
- Also called 'user mode'



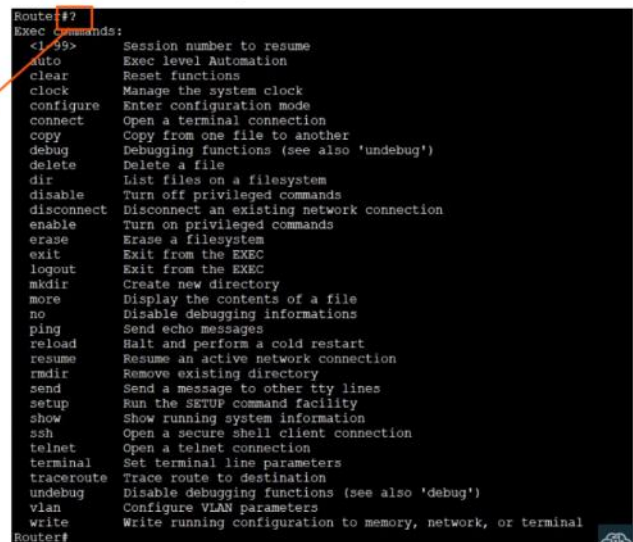
- Provides complete access to view the device's configuration, restart the device, etc.
- Cannot change the configuration, but can change the time on the device, save the configuration file, etc.

Cisco IOS CLI

User EXEC Mode



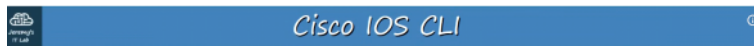
Privileged EXEC Mode



Use a question mark (?) to view the available commands

CLI shortcuts - TAB

15 February 2026 16:06



I pressed Tab here

```
Router>en
Router>enable
Router#
```

```
Router>en
Router>enable
Router#
```

"en" worked because there's only one command that starts with "en"

```
Router>en
Router#
```

```
Router>en
Router#
```



```
Router>e
% Ambiguous command: "e"
Router>
```

```
Router>e
% Ambiguous command: "e"
Router>
```

When you type "e", the device shows "ambiguous command."
This means there is more than one command that starts with the letter "e."

```
Router>e
% Ambiguous command: "e"
Router>e?
enable exit
Router>e
```

```
Router>e?
enable exit
Router>e
```



Global configuration Mode

15 February 2026 16:10



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
```

```
Router>enable
Router#con?
configure connect
Router#conf t?
terminal
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
```

You can just type conf t as a shortcut for configure terminal.

Enable password

15 February 2026 16:48

```
enable password

Router(config)#enable password?
password
Router(config)#enable password ?
 7 Specifies a HIDDEN password will follow
LINE The UNENCRYPTED (cleartext) 'enable' password
level Set exec level password
Router(config)#enable password CCNA ?
<cr>
Router(config)#enable password CCNA
Router(config)#
```

1. enable password?

If you type ? without a space, the CLI shows the possible commands or keywords that match what you have typed so far.

- It is used for **command completion** or to see matching options.

2. enable password ?

If you type a space before ?, the Cisco CLI help feature shows the available parameters or options for that specific command.

Example output:

- 7 → Specifies that a **hidden (encrypted)** password will follow.
- LINE → Sets an **unencrypted (cleartext)** enable password.
- level → Sets the **exec level** password.

You do **not** type LINE.

You only need to type the password you want to set.

3. Command example

```
Router(config)# enable password CCNA
```

- Here, **CCNA** is the **enable password**.

Important note:

Cisco CLI is **case sensitive**, so:

- **CCNA** and **ccna** are **different** passwords.

```
enable password

Router(config)#enable password CCNA
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#exit

Router con0 is now available

Press RETURN to get started.

Router>enable
Password:
Router#
```

```
enable password

Router>enable
Password:
Router#
```

- The password does **not** display as you type it (for security purposes).

```
Router>enable
Password:
Password:
Password:
% Bad secrets

Router>
```




enable password

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable password CCNA
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#exit
```

```
Router>enable
Password:
Router#
```

Config files / show startup-config

15 February 2026 18:13



running-config / startup-config

- There are two separate configuration files kept on the device at once.
- **Running-config** = the current, active configuration file on the device. As you enter commands in the CLI, you edit the active configuration.
- **Startup-config** = the configuration file that will be loaded upon restart of the device.



show running-config / show startup-config

```
Router#show running-config
Building configuration...

Current configuration : 714 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable password CCNA
!
```

```
Router#show startup-config
startup-config is not present
```

Because we haven't saved the running configuration yet, the router does not have a startup configuration.

So, every time the router restarts, it will load the **default configuration** instead of the running configuration.

In simple terms:

- **Running configuration** → current active settings (stored in RAM).
- **Startup configuration** → saved settings (stored in NVRAM).
- If you don't save the running config, it will be lost after a reboot.

```
Router#write
Building configuration...
[OK]
Router#write memory
Building configuration...
[OK]
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

There are three ways to save the running configuration as the startup configuration:

1. write
Saves the current running configuration.
 2. write memory
Saves the running configuration to the startup configuration in memory.
 3. copy running-config startup-config
Copies the running configuration and stores it as the startup configuration.
- All three commands perform the same function—they save the current configuration so it remains after the device is rebooted.

```
Router#show startup-config
Using 714 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable password CCNA
!
!
```

The password is currently visible. To improve security, we will learn how to keep it hidden in the next slide

Service password encryption

15 February 2026 23:11

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#service password-encryption

Router#show running-config
Building configuration...

Current configuration : 719 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable password 7 08026F6028
!
```

service password-encryption

- This Cisco IOS command **encrypts** plain-text passwords (Type 0) in the configuration.
- It converts them into **Type 7 passwords**.
- **Type 7 is weak and reversible**, not meant for real security.
- It is used only to **hide passwords from casual viewing** in the config file.

Better practice:

Use enable secret (Type 5/8/9 hashing) for stronger security.

service password-encryption

Type 7 Password: 08026F6028

Crack Password

Plain text: CCNA

Type 7 encryption:

- Is **very weak**.
- Can be **easily decrypted**.
- Is meant only to **hide passwords from casual viewing**.

enable secret

```
Router(config)#enable secret Cisco
Router(config)#do sh run
Building configuration...

Current configuration : 766 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$YlCkLMcTYWwkF1Cndt1l.
enable password 7 08026F6028
```

5 = MD5 encryption

NOT used

This password is:

- **Hashed** (not just encrypted).
- Stored using a **stronger algorithm**.
- Much more secure than enable password.

If both are configured:

enable password <value>
enable secret <value>

- The router **always uses enable secret**
- enable password is ignored.

Cancelling command

16 February 2026 00:23

cancelling commands

```
Router(config)#no service password-encryption
Router(config)#do sh run
Building configuration...

Current configuration : 769 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$YlCkLMcTYWwkF1Ccndt1l.
enable password 7 08026F6028
!
```

- In Cisco IOS, most commands can be removed by adding the word **no** before them.
- So:
 - service password-encryption → turns on password encryption.
 - no service password-encryption → turns it off.
 - do lets you run privileged EXEC commands while you are in configuration mode.
 - sh run = short for show running-config.
 - So this command displays the current configuration.
- When you disable service password-encryption:
 - Passwords that were already encrypted **will not be decrypted**.
 - But any **new passwords you configure after disabling** it will appear in **plain text** in the configuration.

no service password-encryption — Short Note

- Disables the password encryption feature.
- New passwords will be stored in **plain text (Type 0)**.
- **Already encrypted passwords remain encrypted** (they are not decrypted automatically).

Best practice:

Use enable secret (Type 5/8/9 hashing) for real security.

Review- service password

16 February 2026 03:36



service password-encryption

If you enable service password-encryption...

- *current passwords will be encrypted.*
- *future passwords will be encrypted.*
- *the enable secret will not be effected.*

If you disable service password-encryption...

- *current passwords will not be decrypted.*
- *future passwords will not be encrypted.*
- *the enable secret will not be effected.*

Review

16 February 2026 03:47

Router> = user EXEC mode

Router>**enable**
##used to enter privileged EXEC mode

Router# = privileged EXEC mode

Router#**configure terminal**
##used to enter global configuration mode

Router(config)# = global configuration mode

Router(config)#**enable password** password
##configures a password to protect privileged exec mode

Router(config)#**service password-encryption**
##encrypts the enable password (and other passwords)

Router(config)#**no** command
##removes the command

Router(config)#**enable secret** password
##configures a more secure, always-encrypted enable password

Router(config)#**show running-config**
##displays the current, active configuration file

Router(config)#**run** privileged-exec-level-command
##executes a privileged-exec level command from global configuration mode

Router(config)#**show startup-config**
##displays the saved configuration file which will be loaded if the device is restarted



Command Review

Router(config)#**write**
##saves the configuration

Router(config)#**write memory**
##saves the configuration

Router(config)#**copy running-config startup-config**
##saves the configuration

